Example 1:- Solve the following system of linear equation using "Gauss Elimination" Method:-

$$x + y + z = 6$$
$$2x + 3y + z = 13$$
$$3x + 2y + 5z = 18$$

Forward ~~Elimination~~ Elimination:- Here,

$R_1 : x + y + z = 6$ ——— (1)

$R_2 : 2x + 3y + z = 13$ ——— (2)

$R_3 : 3x + 2y + 5z = 18$ ——— (3)

Step 1:- Remove x from R_2

Pivot Row : R_1

Elimination Row : R_2

$$R_2' = R_2 + \left(\frac{-2}{1}\right)R_1$$
$$= R_2 - 2R_1$$

$$\boxed{\text{Pivot} = -\frac{R_2\text{'s coefficient of } x}{R_1\text{'s coefficient of } x}}$$

$$\begin{array}{r} 2x + 3y + z = 13 \\ -2x - 2y - 2z = -12 \\ \hline y - z = 1 \end{array}$$ ——— (2')

Step 2: Eliminate x from R_3

Pivot Row = R_1

Elimination Row = R_3

$$R_3' = R_3 + \left(\frac{-3}{1}\right) R_1$$

$$= R_3 - 3R_1$$

$$\begin{array}{r} 3x + 2y + 5z = 18 \\ -3x - 3y - 3z = -18 \\ \hline -y + 2z = 0 \quad \text{—— (3')} \end{array}$$

Now the system becomes

$R_1 : x + y + z = 6$ —— (1)

$R_2' : y - z = 1$ —— (2')

$R_3' : -y + 2z = 0$ —— (3')

Step 3: Eliminate y from R_3'

Pivot Row = R_2'

Elimination Row = R_3'

$$\therefore R_3'' = R_3' + \left(-\frac{-1}{1}\right) R_2'$$

$$= R_3' + R_2'$$

$$\begin{array}{r} -y + 2z = 0 \\ y - z = 1 \\ \hline z = 1 \quad \text{—— (3'')} \end{array}$$

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Back Substitution :-

Now, the system becomes

$$x + y + z = 6 \text{ —— } (1)$$
$$y - z = 1 \text{ —— } (2')$$
$$z = 1 \text{ —— } (3'')$$

from 3'',

$z = 1$

from 2',

$y = 1 + z$

$= 1 + 1$

$= 2$

from 1,

$x = 6 - y - z$

$= 6 - 2 - 1$

$= 3$

$\therefore x = 3,\ y = 2,\ z = 1$

Example 2:- Solve the following system of linear equations using Gauss Elimination with Partial Pivoting

$$3x + 5y + z = 0$$
$$3x + 2y + 5z = 10$$
$$6x + 4y + 2z = 12$$

Solution: Forward Elimination :-

~~Step 1:~~

$R_1 : 3x + 5y + z = 0$ —— (1)

$R_2 : 3x + 2y + 5z = 10$ —— (2)

$R_3 : 6x + 4y + 2z = 12$ —— (3)

Step 1:- Find the largest coefficent of the 1st column

$R_3 \longleftrightarrow R_1$

> Note: While finding the largest coefficient, only consider the magnitude. So, think of it as an absolute value

Now, the system becomes,

$R_1 : 6x + 4y + 2z = 12$

$R_2 : 3x + 2y + 5z = 10$

$R_3 : 3x + 5y + z = 0$

Step 2: Remove x from R_2

$$R_2' = R_2 + \left(-\frac{3}{6}\right) R_1$$
$$= R_2 - \frac{1}{2} R_1$$

$$
\begin{array}{r}
3x + 2y + 5z = 10 \\
-3x - 2y - z = -6 \\
\hline
4z = 4 \quad \text{——} (2')
\end{array}
$$

Step 2: Remove x from R_3

$R_3' = R_3 + \left(-\frac{1}{2}\right) R_1$

$= R_3 - \frac{1}{2} R_1$

$$
\begin{array}{r}
3x + 5y + z = 0 \\
-3x - 2y - z = -6 \\
\hline
3y \quad = -6 \quad \text{——} (3')
\end{array}
$$

Now the system becomes,

$R_1: \quad 6x + 4y + 2z = 12$ —— (1)

$R_2': \quad 0 + 4z = 4$ —— (2')

$R_3': \quad 3y + 0 = -6$ —— (3')

Step 3: Swapping R_3' with R_2' ($R_3' \longleftrightarrow R_2'$), system becom

$R_1: 6x + 4y + 2z = 12$ —— (1)

$R_2': \quad 3y + 0 = -6$ —— (2')

$R_3': \quad 0 + 4z = 4$ —— (3')

Note: Here if coefficient of y was not already 0 for R_3', we would needed to eliminate it ourselves by considering R_2' as pivot row

Back Substitution:-

from 3' we get,

$$4z = 4$$

$$\Rightarrow z = 1$$

from 2' we get,

$$3y = -6$$

$$\Rightarrow y = -2$$

from 1 we get,

$$6x + 4y + 2z = 12$$

$$\Rightarrow 6x = 12 - (4y + 2z)$$

$$\Rightarrow 6x = 12 - (-8 + 2)$$

$$\Rightarrow 6x = 12 + 6$$

$$\Rightarrow x = \frac{18}{6} = 3$$

$\therefore x = 3, \; y = -2, \; z = 1$